A THICKENING OF ALGEBRAIC SPACES INDUCES AN EQUIVALENCE OF SMALL ETALE SITES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1816-1825 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 1827-1834, the complete author proof (6 body lines).

Editorial note. Navigation stubs: 2 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The topological invariance theorem for big etale sites of a thickening (same chapter, collected as S3-spaces-topological-invariance L1411) and the scheme-preservation lemma lemma-thickening-scheme (this batch, L16042); cited inputs with the restriction argument written in the retained span.

Source reference. Original lemma 05ZR: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1730–1733. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05ZH: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 1316–1326. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $X \subset X'$ be a thickening of algebraic spaces over S. The functor*

$$V' \longmapsto V = X \times_{X'} V'$$

defines an equivalence of categories $X'_{\acute{e}tale} \to X_{\acute{e}tale}$.

Proof. The functor $V' \mapsto V$ defines an equivalence of categories $X'_{spaces,\acute{e}tale} \to X_{spaces,\acute{e}tale}$, see Theorem 05ZH. Thus it suffices to show that V is a scheme if and only if V' is a scheme. This is the content of Lemma 05ZR. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.